ChArUco composite calibration target

DATASHEET Rev A · 2026-10-24

Overview

The ChArUco composite calibration target combines a checkerboard grid with ArUco fiducial markers on a rigid aluminium composite substrate. It enables simultaneous intrinsic and extrinsic calibration with sub-pixel corner detection and unique marker identification.

Suitable for industrial machine vision, robotics, and research applications where a flat, dimensionally stable target is required.

Available sizes

Part number	Board size	Grid	Square	Marker	Dictionary	Tolerance
CHA-P-6X6-200x150-25	200 x 150 mm	7 x 10	25 mm	18 mm	DICT_6X6_250	+/-0.3 mm
CHA-P-6X6-300x200-25	300 x 200 mm	7 x 10	25 mm	18 mm	DICT_6X6_250	+/-0.3 mm
CHA-P-6X6-400x300-30	400 x 300 mm	9 x 12	30 mm	22 mm	DICT_6X6_250	+/-0.3 mm
CHA-P-6X6-600x400-30	600 x 400 mm	12 x 18	30 mm	22 mm	DICT_6X6_250	+/-0.3 mm
CHA-P-6X6-800x600-40	800 x 600 mm	14 x 18	40 mm	30 mm	DICT_6X6_250	+/-0.3 mm

Substrate

- Material: aluminium / LDPE composite (Dibond-type), 6 mm thick
- Surface: UV-printed, matte finish, scratch-resistant
- CTE: approximately 24 um/(m.K)
- Operating temperature: -50 to +80 C
- Weight: 400 g (200 x 150 mm) to 3200 g (800 x 600 mm)

Feature accuracy

- Feature tolerance: +/-0.3 mm + 0.2 mm/m (board-size dependent)
- Flatness: <= 0.5 mm/m
- Corner radius: 6 mm

Pattern details

The ChArUco pattern consists of a checkerboard where the white squares contain ArUco markers from the specified dictionary. Corner detection uses the checkerboard intersections; marker IDs provide unique identification and allow partial board visibility.

Default dictionary: DICT_6X6_250 (250 unique markers, 6x6 bit grid).

Custom dictionaries (DICT_4X4_50, DICT_5X5_100, AprilTag 36h11) available on request.

Mounting

Four M4 threaded inserts are installed in the rear panel for tripod or fixture mounting. A 1/4-20 UNC fixture adaptor is available separately (ACC-A-MNT-01).

Deliverables

Each target ships with:

- Dimensional inspection report (included)
- Datasheet PDF
- DXF board outline
- OpenCV JSON configuration
- Kalibr YAML configuration
- ROS 2 YAML configuration
- HALCON description file

ILAC-recognised calibration certificate available as an add-on (SVC-X-CERT-ILAC).

Storage and handling

Store flat, away from direct sunlight and moisture. Do not stack heavy items on the target. Clean with a soft, dry cloth. Avoid solvents.

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